

Particular Specification

BUILDING INFORMATION MODELLING (BIM)

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PARTICULAR SPECIFICATION

BUILDING INFORMATION MODELLING (BIM)

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The Contractor is required to apply Building Information Modelling (BIM) technology for this project according to the design concept for the Works, provide the services and submit all the deliverables as detailed in this Particular Specification.
- B. The Contractor is required to set up a BIM Team with a Building Information Modelling Manager (BIM Manager) leading the BIM Team and to provide, a minimum of, disciplinary BIM coordinators and modellers in architectural, structural and building services disciplines as detailed in the below table. Qualifications of the BIM Team are specified in Appendix 1 of this Particular Specification.

	Disciplinary BIM Coordinator (No.)	Modeller (No.)
Architectural	1	1
Structural		1
Building Services	1	2

1. The BIM Team Leader of the Contractor shall carry out the following duties and functions
- a. Lead and manage the BIM Team and shall be responsible for the overall BIM management and provide control to ensure the deliverables are issued on time,
 - b. Coordinate with the Architect, Structural Engineer and Building Services Engineer,
 - c. Plan, develop and manage the project BIM standard and implementation strategy,
 - d. Provide quality control and checking procedures,
 - e. Ensure all the deliverables are issued on time,
 - f. Provide advice/guidance to the Architect and Engineers in
 - 1) Reviewing BIM modelling formats/protocol and co-ordination,
 - 2) Reviewing analysis and findings on application by using different software's such as the clash detection and resolving the clashes,
 - 3) Providing technical support to the project team regarding modelling enquiries.

2. The BIM Team shall provide the following services:
 - a. To coordinate all parties including but not limited to different design disciplines, sub-contractors, Specialist Sub-contractors, Specialist Contractors, suppliers, Client/architect and his/her representatives, Government Departments and utility undertakings and collecting suitable information and data from them,
 - b. To build and develop the BIM models and all deliverables based on the information and data collected, to manage the BIM databases, to report any clash/conflict or difficulties in BIM models production and to resolve such difficulties with relevant parties,
 - c. To utilise BIM for avoidance of risks and minimisation of changes, to monitor construction progress in various stages, and to enhance safety design for construction and operation,
 - d. To input the updated information and data, including but not limited to all variations, from time to time; to create, manage, develop and update the BIM models to all deliverables for the construction works; to ensure the updated BIM models are coordinated and always kept ahead of the site construction to enhance the project delivery process through BIM adoption,
 - e. To coordinate the BIM models and all deliverables to ensure consistency among all deliverables; to use the best practice in setting up the system for coordinating BIM models among all parties,
 - f. To provide technical support and guidance to the Client/architect and his/her representatives in using the BIM models, managing the BIM database and resolving potential construction difficulties,
 - g. To consolidate design information from the Sub-contractor(s) and develop the BIM model and all deliverables as required in this Contract.
3. The Contractor shall submit the organization chart and qualification of the members of the BIM Team for the Architect's approval within 7 calendar days after the commencement of the Contract. For any proposed staff movement or change in the BIM Team, the Contractor shall notify the Architect as soon as possible and provide a CV of the replacement personnel together with evidence of equivalent BIM competency to the Architect within 7 calendar days of the notification for approval. The Contractor shall provide sufficient and technically competent resources as agreed or directed by the Architect or his/her representatives in order to complete all BIM tasks and deliverables specified in the Contract according to the approved programme.

1.02 REFERENCES

The BIM Manager should adopt the predominant BIM industry standard on discharging his duties. Reference may be made to the published guidelines, such as:

1. CIC BIM Standards - General (Version 2.1 - 2021), with new templates in Appendix D, by the Construction Industry Council in Hong Kong;
2. CIC BIM Exchange Information Requirements (EIR) Template (BIM Specifications) (Version 1.1 – 2021), by the Construction Industry Council in Hong Kong;
3. CIC BIM Standards Architecture and Structural Engineering (Version 2.1 - 2021), by the Construction Industry Council in Hong Kong;
4. CIC BIM Standards for Mechanical, Electrical and Plumbing (Version 2 – 2021), by the Construction Industry Council in Hong Kong ;
5. CIC BIM Standards for Underground Utilities (Version 2 – 2021), by the Construction Industry Council in Hong Kong ;

6. CIC Production of BIM Object Guide - General Requirements (2021), by the Construction Industry Council in Hong Kong ;
7. AEC (UK) BIM Technology Protocol – Practical implementation of BIM for the Architectural, Engineering and Construction (AEC) industry, version 2.1.1, June 2015, by the AEC (UK);
8. BIM Project Execution Planning Guide, version 3.0, March 2021, by The Computer Integration Construction Program of the Pennsylvania State University;
9. BIM Project Specification, revision 3.0, Jun 2011, by the Hong Kong Institute of Building Information Modelling;
10. Building Information Modelling for Asset Management (BIM-AM) Standards and Guidelines, version 2.0, 2019 issued by the Electrical and Mechanical Services Department;
11. Building Information Modelling (BIM) Guide for Building Services Installation (Version 1.1) issued by Building Services Branch (BSB), Architectural Services Department;
12. Building Information Modelling (BIM) Guide for Cost Estimation (Version 1.0) issued by Quantity Surveying Branch (QSB), Architectural Services Department;
13. Building Information Modelling (BIM) Guide for Facilities Upkeep (Version 1.2) issued by Property Services Branch (PSB), Architectural Services Department;
14. Drafting Specification for Engineering Survey, Rev. 3.0/Nov 2014 by Civil Engineering and Development Department;
15. BS EN ISO 19650-1:2018 Organization and digitization of information about buildings and civil engineering works, including building information modelling (BIM) – Information management using building information modelling, Part 1: Concepts and principles;
16. BS EN ISO 19650-2:2018 Organization and digitization of information about buildings and civil engineering works, including building information modelling (BIM) – Information management using building information modelling, Part 2: Delivery phase of the assets;
17. PAS 1192-3:2014 Specification for information management for the operational phase of construction projects using building information modelling;
18. PAS 1192-5:2015 Specification for security-minded building information modelling, digital built environments and smart asset management;
19. BS 1192-4:2014 Collaborative production of information. Fulfilling employer's information exchange requirements using COBie. Code of practice.

1.03 PERFORMANCE REQUIREMENTS / OBJECTIVES

The objective of the application of BIM is to create a digital building information model for the project and to use BIM as a platform to facilitate project planning, site administration, safety planning, design co-ordination, clash detection prior to construction, minimization of abortive works, waste reduction and efficient asset management in order to achieve the following beneficial purpose:

1. To minimize design discrepancies, improve design co-ordination and deliver a clash-free design through the use of 3D modelling technique,
2. To enhance visual communication between the Contractor and all stakeholders, improve mutual understanding of the design intent and facilitate design review and vetting process,
3. To deliver drawings, including Building Service Drawings, Combined Services Drawings (CSDs) and Combined Builder's Work Drawings (CBWDs),

4. To support the development of 4D modelling construction sequence during the construction stage to enhance communication, predict and manage construction process,
5. To support the development of asset management by using BIM with an as-built BIM model for effective operation and maintenance of the buildings.

1.04 DELIVERABLES

The Contractor shall submit the following deliverables according to the programme stipulated in this Particular Specification.

Except otherwise stated in the Conditions of Contract, the copyright of all data and information contained in all deliverables including BIM models and reports created, developed and provided under this Contract, including all draft and final versions, shall be the property of the Employer.

A. BIM Project Execution Plan:

The Contractor shall prepare and submit a BIM Project Execution Plan as per **Appendix 2** for the Architect's approval.

The Contractor shall define the full details of the implementation and collaboration process in the BIM Project Execution Plan.

The Contractor shall review the BIM Project Execution Plan regularly, and update as necessary or upon request by the Client/architect and his/her representatives, for the purpose of enhancement in project benefit.

B. Design Authoring and Design Reviews

The Contractor shall create, maintain and update the BIM models, including the following:-

1. A Base Model modelled according to Contract Documents for both 3D geometry and information, which serves as a basis for contract administration;
2. Updated model for each Architect's instruction.

The Contractor shall use BIM models to show the following components to the agreement and satisfaction of the Architect or the Engineer before construction, but not limited to:

3. Architectural BIM model

In construction stage, model elements / BIM objects shall be modelled at minimum Level of Information Need (LOIN) 300 in general. For AM and facilities upkeep purpose, model elements / BIM objects shall be field verified representations in terms of size, shape, location, quantity, and orientation.

Model elements / BIM objects to be built shall meet and satisfy the needs of the required BIM Uses as specified in the Scope of Works, Guidelines and Standards fit for purposes. Sufficient LOIN shall be applied.

In Construction stage, LOD-G of model elements shall comply with the followings:

<u>Discipline</u>	<u>LOD-G</u>
Architectural	300
Structural	300
MEP	400

During construction, all model elements shall be field verified progressively in a Digital Works Supervision System (DWSS).

In terms of LOD-I, the list of elements, include but not limited to, in the table below specifying the required in various stages.

Abbreviation of progress stages

CT Construction

AB As Built

<u>Architectural components</u>	<u>LOD-I</u>	
	CT	AB
1. Existing site topography, services and buildings; including access to site;	200	200
2. Site context of surrounding;	200	200
3. Generic models for design options;	300	300
4. Rooms, spaces, corridors, plant & equipment rooms;	300	300
5. Floor slabs, ramps & roofs;	400	500
6. Internal and external walls and columns;	400	500
7. Doors, shutters and their hoods, access panels;	400	500
8. Ironmongery;	400	500
9. Curtain wall & precast facades;	400	500
10. Windows and louvers;	400	500
11. Lift doors;	400	500
12. Balustrades, parapets and railings;	400	500
13. Staircases;	400	500
14. Ramps, raised access floor, false ceilings with access openings;	400	500
15. Built-in fixed furniture such as toilet partitions, cabinets;	400	500
16. Smoke barriers;	400	500
17. Drainage/ Services channel covers;	400	500
18. Cat ladders, catwalks, maintenance platforms;	400	500
19. Exterior elements such as canopies, sun-shading devices, wall features;	400	500
20. Sanitary Fitments;	400	500
21. Signage;	400	500
22. External Works such as soft landscape, hard landscape, etc,	400	500

4. Structural BIM model

<u>Structural components</u>	<u>LOD-I</u>	
	CT	AB
1. All elements requiring structural, foundation and geotechnical design;	400	500
2. Any precast/ prefabricated & prestressed elements;	400	500
3. Details required in congested areas, cast-ins, changes in level and junctions/ connections between structural elements;	400	500
4. All penetrations /interfaces between structural members and building services/ E&M services /builder's works. (e.g. openings, curbs, voids, pits, recesses, mass concrete, etc);	400	500
5. Reinforcement details at certain congested zones, to be selected by the Engineer or his representatives, for interfacing coordination between reinforcement and building services elements;	400	500
6. Specific rendered views, to be selected, from the BIM model on the structural framework and interfaces to illustrate connections among structural members;	N/A	N/A

5. Building Services BIM model

<u>MEP components</u>	<u>LOD-I</u>	
	CT	AB
1. All equipment to maintain the building operation, including backup system	400	500
2. All equipment for system monitoring, e.g. sensor, visual surveillance	400	500

The building services components/objects shall also refer to the Building Information Modelling for Asset Management (BIM-AM) Standards and Guidelines, version 2.0, 2019 issued by the Electrical and Mechanical Services Department.

6. For efficient handling of models, the BIM model should be sub-divided into separate zones/ services/ systems as appropriate to maintain reasonable file size of the models. The model sub-division strategy (by zones/ services/ systems) should be stated in the BIM Project Execution Plan. File sizes of each sub-divided BIM model shall be kept in minimum by purging of unused views, BIM objects and settings before publish or submission. The maximum file size for each sub-divided BIM model should not exceed 500MB unless otherwise approved.

C. Existing Conditions Modelling

The Contractor shall make use of 3D digital survey technology to provide existing conditions model and as-built condition verified for the Works. It provides documentation of environment for future modelling and 3D design coordination. It can supplement photographic records of condition survey. The 3D digital survey model shall be georeferenced to the same absolute coordinate system, e.g. Hong Kong 1980 Grid, according to x, y, z coordinates. The format can be:

1. Point Cloud (.las and .rcs)
2. Build mesh (.tin)
3. Export ortho-image (.jpeg, .png and .tiff)
4. Video (H.264 & other formats specified in the latest version of OGCI0 – The HKSARG Interoperability Framework (S18))

D. 3D Coordination

1. The Contractor shall carry out clash analysis weekly during the construction period based on the different BIM models created and updated at different stages in order to resolve all major system clashes prior to construction by identifying and resolving clashes among different disciplines. However, Clash Analysis shall NOT be considered as the only method of quality check for spatial design coordination.
2. Clash analysis shall include the checking of headroom requirements and working spaces for building services operations and maintenance activities. In particular, for major equipment which must provide services without any break, dynamic envelope model showing the estimated dimensions of the physical space sufficient for equipment delivery and maintenance is required as well as installation sequence. The Contractor shall liaise with the Architect or the Engineer to determine the criteria level of acceptable clashes, e.g. sprinkler pipes pass through beams.
 - a. Compare BIM models built up from design of different disciplines and shop drawings from sub-contractors
 - b. Identify and visualize clashes
 - c. Report to the Architect / Engineer or his/her representatives in an agreed format
 - d. revise design and shop drawings
 - e. Revise BIM models
 - f. Perform another round of analysis until clashes are resolved.
3. Clash analysis should be presented in the form of a report to compare the different clashes, record the clash detection process and assumptions on element tolerances, areas and elements, identify any major conflict discovered in the process and generate resolution result summary. The report shall address the following.
 - a. Software used
 - b. Process overview
 - c. Responsibilities
 - d. Outputs
 - e. Technical query workflow
 - f. Clash resolution process
 - g. Action plan with target completion schedule to handle and resolve detected clashes

- h. Tolerance levels (mm) for different discipline
- i. Operation clearance
- j. Maintenance clearance
- k. Buildability
- l. Services compatibility

E. Phase Planning (4D Modelling)

1. The Contractor shall provide 4D Model simulations to the construction process of the Works to:
 - a. Establish relationships between the programme and sequence of construction activities including the delivery of material and equipment to be carried out during the construction,
 - b. Demonstrate the Contractor's works sequences,
 - c. Identify potential time and spatial conflicts,
 - d. Optimize the use of critical resources,
 - e. Enhance safety requirements and construction process control,
 - f. Minimize disturbance to the neighbourhood,
 - g. Better co-ordinate with affected parties and resolve interfacing issues at early stages,
 - h. Monitor procurement status of project materials.
2. The 4D simulations shall be demonstrated in monthly intervals linking all activities in the master programme and it shall be automatically matched with the activities as shown in the master programme with appropriate file format.
3. 4D Simulation Report

The Contractor shall prepare a 4D programme for construction progress monitoring and demonstrated in monthly intervals, linking all activities in the master programme and it shall be automatically matched with the activities as shown in the master programme with appropriate file format. The 4D programme shall evolve with project progression. Time and other 4D-related information within Work-in-progress (WIP) BIM shall be concurrent with the outputs. The 4D programme shall be used in the project progress meeting, at a monthly interval, to indicate and compare the current actual construction progress with the baseline programme to facilitate project management.

The Contractor shall submit a 4D Simulation Report containing the following but not limited to:

- a. Description of the 4D simulation report, e.g., assumptions, time interval, construction method statement, guide for accessing the files and models...etc.
- b. Video(s) of the 4D simulation.
- c. The 4D walkthrough/ flythrough simulation/ animation videos from the BIM shall be minimum [1080p] resolution, [30] frames per second, and within [2] minutes each. The videos of the 4D simulations shall be submitted in required file format viewable in standalone free viewer.
- d. BIM native model(s).
- e. Models(s) for 4D simulation platform.
- f. Linked project programme or equivalent deliverable.

F. Digital Fabrication

For mass customized components which are of large quantities and variety in dimension, shapes, geometries, etc. and modular construction units, the contractor shall digitalize the construction details in BIM model to facilitate the fabrication of construction materials.

G. Site Utilization Planning

The Contractor shall use the 4D model by assigning the transportation path of the construction plant and equipment in the model to demonstrate the feasibility and effectiveness of the site logistic arrangements. For construction activities with very high to extreme risk level identified from the Systematic Risk Management (SRM) according to ETWB TC(W) No. 6/2005 or other identified activities of significant added values at construction stage, all temporary works and the Contractor's proposed method statements of the Works shall be modelled to demonstrate the feasibility and to prove the constructability and buildability of the proposed method statement.

H. As-built Modelling

1. The Contractor shall prepare and submit the as-built BIM models of all components (including Architectural, Structural and Building Services Installation) as described in this Particular Specification.
2. The as-built BIM models shall be based on the final approved construction information that had actually been built, and shall be used to produce as-built drawings. Information on location such as room number and building name, staircase number, washroom number, lift lobby number is required to be incorporated into the as-built BIM models. The Contractor shall also import and display the operation data, product catalogues, manuals, warranties and maintenance history of equipment etc. into the as-built BIM models. The as-built BIM models shall be able to create sheet records.
3. The Contractor shall include as-built construction and equipment components of the Works with information as listed below for approval:
 - a. Room Data Sheets;
 - b. Door Schedules;
 - c. Ironmongery Schedules;
 - d. Window Schedules;
 - e. Access Panel Schedules;
 - f. Shutter Schedules;
 - g. Cat ladder Schedules;
 - h. Louvre Schedules;
 - i. Sanitary Fitment Schedules;
 - j. Signage Schedules;
 - k. Manhole Schedule;
 - l. Lighting Schedule;
 - m. Roofing System;
 - n. Comprehensive materials data sheet list and completed materials / equipment warranty list;
 - o. As-built BIM models and 2D drawing files;
 - p. Export data files;
 - q. Folder storing all the object files;

- r. Testing and Commissioning reports;
- s. Operation and Maintenance manuals;
- t. Training Plan;
- u. Training Manual;
- v. Maintenance Plan;
- w. Relevant statutory certificates, approval documents and forms; and
- x. Other relevant project information as required.

The as-built BIM models and 2D as-built record drawings shall comprise customized building attributes and file structure for data submission in format agreed and approved by the Architect. The as-built information format, structure, data arrangement and other details shall also be proposed and submitted to the Architect for approval.

- 4. The required as-built data and relevant documentations shall be stored in a standardized file folder structure.
- 5. The Contractor shall consolidate the COBie/IFC/asset information requirements and other O&M information requirements from the users and prepare a proposal of COBie/IFC/asset information requirements for acceptance by the Architect and Engineer.
- 6. For as-built model, use laser scanning for field verification method and for verifying the as-built elements of MEP if necessary.
- 7. The Contractor shall make reference to EMSD's BIM-AM Standards and Guidelines (Version 2.0) in delivering the as-built BIM model for MEP installation.
- 8. Maintenance Scheduling

The Contractor shall provide and input the following information or documents related to building fabric and structure, subject to periodic maintenance inspection or testing into the as-built BIM models:

- a. Recommended frequency of maintenance inspection or testing;
- b. Recommended scope of maintenance inspection or testing activities;
- c. Recommended maintenance cycle of the building fabric or structure;
- d. Shop drawings;
- e. Operation and Maintenance Manuals;
- f. Warranties and guarantees (together with the names and addresses of specialist contractors and/ or suppliers);
- g. Technical information (such as specifications of materials and design standards); and
- h. Relevant material certificates and test reports, etc.

I. Drawing Generation (Drawing Production)

- 1. The Contractor shall generate drawings in the required file format from the BIM models including but not limited to architectural drawings, building services drawings, structural drawings, Combined Services Drawings, Combined Builder's Works Drawings etc. to facilitate the co-ordination and operation for the construction of the Works during the contract period. As far as practicable, the 2D drawings should be generated from the BIM authoring software directly.
- 2. It is acceptable that certain 2D drawings such as architectural details, building services schematic /control logic diagrams /drawings and reinforcement details cannot be generated from the BIM model directly. Alternative software, i.e. non-BIM authoring software, could be used and these 2D drawings shall be prepared in accordance with the CAD Standard for Works Projects (CSWP).

J. 3D Control and Planning

1. The Contractor shall utilize the BIM model to layout building elements, such as the position of walls using a total station with survey points preassigned in the model; or to automate control of equipment's movement and location, such as using GPS coordinates to determine if proper excavation depth is reached.
2. The Contractor shall conduct virtual mock-up or prototyping for the following areas with an aim to review the effectiveness of spatial design:
 - a. Congested plant & equipment room for installation & future maintenance;
 - b. External cladding system, eg. aluminum claddings and demountable louvers;
 - c. Corridor section with BS installation

The Contractor shall liaise with the Architect / Engineer or his/her representatives to determine the extent of such design reviews.

K. Infrastructures of Asset Management for Building Services Installation

1. The Contractor shall follow the Chapter 5 of the Building Information Modelling for Asset Management (BIM-AM) Standards and Guidelines, version 2.0, 2019 issued by the Electrical and Mechanical Services Department to install the necessary radio-frequency identification (RFID) tags and QR code labels to facilitate the interface of asset identification to the BIM-AM system by the maintenance party.
2. The requirement of "Panel Schedule for Distribution Boards" as stipulated in the EMSD BIM-AM Standards and Guidelines shall be included in the requirements of as-built BIM model for building services installation.

1.05 PROGRAMME

The BIM models are built and developed for the purposes of design co-ordination, phase planning and communication.

The Contractor shall produce and submit at regular intervals updated BIM models on part or whole of the Works in the required file format, such that the project team can easily view and capture the images and perform simple marking-up of annotations on the selected images and transmit through the internet or intranet.

The target programme for production of the BIM models and the deliverables, which will be subject to change according to actual circumstances as per the direction of the Architect or the Engineer, is at **Appendix 3**.

1.06 QUALITY ASSURANCE / CONTROL REQUIREMENTS

A. Quality Assurance / Control Requirements

The Contractor shall establish quality assurance plan for BIM, to ensure appropriate checks on information and data accuracy. The Contractor will be responsible for ensuring the integrity of their BIM and drawings as well as compliance with the BIM standards. The final BIM version shall be fully coordinated and without any conflict with as-built asset attributes.

The quality assurance and control shall include but not limited to the following contents:

1. Model compliance according to the modelling methodology which is stated in the BIM Project Execution Plan
2. Model quality with the LOD and design

3. Division of Worksets and Families
4. Data validation
5. Clash analysis checking
6. As-built point cloud scanning for verification

B. Model Compliance Check

Model compliance checks shall be carried out covering the following:

1. Format, such as software version and extension;
2. Naming, such as naming of the files and their corresponding folders;
3. General settings, such as grid, survey point, project base point, shared coordinate and coordinate system, shared parameters, attributes;
4. Consistency of 2D information generated from model;
5. Attributes for asset entries tracking;
6. Model cleanliness including flag links, unpurged elements and unused views in final model submission;

C. Documentation Compliance Check

Documentation compliance checks shall be carried out to the BIM Project Execution Plans, federation maps, lists of self-check items, clash reports and model register list.

1.07 HARDWARE AND SOFTWARE REQUIREMENTS

- A. The Contractor shall submit a proposal with details on specification and quantities of compatible software and hardware to build up a Common Data Environment (CDE). After the Architect's or the Engineer's approval of the proposal, the Contractor shall provide, set up and maintain the proposed hardware and software for the CDE.
- B. The Contractor shall provide a dedicated CDE for storage, viewing and sharing of BIM models, drawings, animation, rendering and other related files of the Works with the following properties:
 1. It shall have a clear folder structure, being part of the CDE to store various BIM related information,
 2. It shall contain encryption function for data security and be of sufficient capacity to store all files during the whole project life cycle,
 3. It shall be installed with anti-virus software and maintained with updated security patches for all software.
- C. The Contractor shall be required to:
 1. Submit the folder structure of the BIM storage CDE for the Architect's or the Engineer's agreement before uploading files into the BIM storage CDE,
 2. Prepare access control plan of the BIM storage CDE for the Architect's or the Engineer's agreement and ensure no unauthorized access to the BIM storage CDE,
 3. Provide daily backup and off-site backup for the BIM storage CDE in which the backup media shall be stored properly away from the working office,
 4. Maintain the BIM storage CDE to ensure it operates properly during the whole project life cycle,

5. Handover the CDE's files in an approved folder structure format to the Architect before the issuance of the maintenance certificate.

D. Preferred File Formats

The Contractor shall submit native and editable BIM and an open format file of the BIM such as Industry Foundation Classes (IFC) format to the Architect. The Contractor shall ensure that data in the file is in order, without data loss and consistent with those in the native file. The Contractor shall ensure all the BIM deliverables comply with the approved software versions during the contract period and at the time of delivery. The Contractor shall also convert the final version of the BIM to other additional software versions specified by the Architect.

- E. All tools and software applications used must be IFC compliant in order to allow BIM model interoperability.

Qualifications of the BIM Team

Position	Qualification
BIM Manager	1. A valid CIC certified BIM Manager (CCBM) or satisfy the requirements 2, 3 and 4 below; 2. Shall either have corporate membership of an appropriate professional institution or shall have a minimum of five years relevant post-qualification experience plus university degree or equivalent in an appropriate engineering or construction-related discipline; 3. Shall have a minimum of three years of practical experience in management of BIM project; 4. The Contractor shall provide documentary evidence showing that CIC is processing the application of CIC certified BIM Manager form of the proposed "BIM Manager" of the Contractor and within 6 months upon award of the contract, the proposed "BIM Manager" of the Contractor shall obtain and submit the valid certificate of CIC certified BIM Manager to the CIC.
Disciplinary BIM Coordinator:	1. A valid CIC certified BIM Coordinator (CCBC) or satisfy the requirements 2 and 3 below; 2. Minimum 3 years related construction project experience; and 3. Minimum 1 year practical experience in BIM projects
Modellers: Architectural / Structural/ Building Services	1. Diploma holder in construction related discipline; and 2. Minimum 1 year practical experience in BIM projects 3. Minimum 1 year construction related project experience

BIM Project Execution Plan

The Project Execution Plan shall include but not limited to the following sections:

1. BIM Project Execution Plan Overview
2. Project Information
3. BIM Requirements
 - 3.01 BIM Goals
 - 3.02 BIM Uses
 - 3.03 BIM Data
 - 3.04 LOD Responsibility Matrix
 - 3.05 Meeting Schedule
4. BIM Management
 - 4.01 Roles, responsibilities and authority
 - 4.02 BIM Team Resources, Competency & Training
 - 4.03 BIM Deliverable Schedule (Programme)
5. BIM Process
 - 5.01 Common Data Environment (CDE)
 - 5.02 Individual Discipline Modelling
 - 5.03 BIM Coordination and Clash Detection
 - 5.04 Drawing Production
 - 5.05 Model Archive
 - 5.06 Quality Control and Quality Assurance

6. BIM Procedures

- 6.01 BIM Origin Point & Orientation
- 6.02 Modelling Methodology
- 6.03 Model Division
- 6.04 Model Units
- 6.05 Naming Convention
- 6.06 Drawing Sheet Templates
- 6.07 Annotation, dimensions, abbreviation and symbols

7. IT Hardware & Software Solutions

- 7.01 Software Versions
- 7.02 Exchange Formats
- 7.03 Data Security & Back-up
- 7.04 Hardware Specifications
- 7.05 IT Upgrades

8. Asset Management

- 8.01 Requirements and Data Structure of As-built BIM Models
- 8.02 Deliverables and Training
- 8.03 Handover of As-built BIM Models

APPENDIX 3

Programme

Deliverables		Completion Time	
1.0	BIM Project Execution Plan as per Appendix 2.	Within 2 weeks from the date of commencement of the Works	
2.0	i. Architectural and structural initial BIM models in Level of Development together with required 2D drawings (i.e. plans, elevations and sections). ii. Existing Condition Model (refer to 1.04C)	Within 3 weeks from the date of commencement of the Works.	
3.0	Building services initial BIM models together with required 2D drawings (i.e. schematic, layout plans, elevations and sections).	Within 6 weeks from the date for commencement of the Works.	
4.0	Submission of CSD and CBWD BIM models together with required 2D drawings (i.e. plans, reflected ceiling plans, elevations and sections).	Submission by floors within the following listed time frame from the date of commencement of the Works.	
		B2/F – B1/F G/F – 2/F 3/F – 6/F 7/F – 10/F 11/F – 15/F 16/F – UR/F	8 weeks 10 weeks 13 weeks 16 weeks 19 weeks 22 weeks
5.0	Clash Analysis Report	Monthly submission	
6.0	Phase Planning (4D Modelling) and Site Utilization Planning	Monthly submission	
7.0	As-built BIM models and Common Data Environment (CDE)	A draft as-built BIM model shall be submitted at least 1.5 months before completion of the project. The final as-built BIM model shall be submitted along with other record documents to the Architect prior to site handover in accordance with Clause 5.04 under Specification Preliminaries.	

END OF SECTION